year, which you could have received interest on or otherwise used. You would of course need to be compensated for this or the position is not worth taking.

For financial futures, such as equity index futures and bond futures, the interest rate is the main driver of the term structure shape because there is no physical storage required for hedging; you just need to deliver the cash upfront. Therefore, you see less of a steep curve for financial futures than for deliverable ones. On the other hand, there are commodities with severe storage cost where this is the overwhelming factor in the term structure shape. Natural gas, for instance, is very expensive to store and it therefore tends to show very steep contango.

Backwardation, or downwards-sloping term structure, is less common but not an anomaly. Backwardation can be caused by seasonality, interest rate conditions or unusual storage cost situations and is not uncommon for softs and perishable commodities.

Basis Gaps

The current price of two contracts with the same underlying but different delivery months will always be different and this is reflected in the term structure. The April gold contract will be traded at a different price than the December gold contract for the same year, and the same logic goes for any other market as well. Usually the price of the December contract will be higher than the April contract of the same year, in which case we have a contango situation, and this has nothing to do with the expectations of gold price changes. It may be intuitive to think that the higher price of the December contract reflects traders' view that the gold spot price should move up, but this is not at all the case. Instead the hedging cost, or carry cost if you will, is the core factor in play. The difference in base price between two contracts becomes acutely important when the currently traded contract comes to the end of its life and you need to roll to the next.

Figure 2.4 shows the May and July 2012 rough rice price where the July contract is the lighter, broken line. Note that the July is consistently more expensive than the May. This is the normal case but there are times when the relationship can reverse as well and the longer contract can be cheaper than the shorter.

The reason for this price discrepancy should now be clear: it's primarily related to hedging or carry costs. The problem that this creates for us is that when creating a continuous time series for long-term simulations, we